Net Sack Yards: Measuring Pass-Rush Production Net of Pass-Rush Indiscipline in the 2025 NFL Regular Season

A descriptive study of all 32 teams, 2025 regular season only

Abstract

Sack totals reward a defense for ending plays behind the line of scrimmage but ignore the yardage a defense hands back through careless pass-rush fouls. This paper defines **Net Sack Yards** (sack yards gained minus yards surrendered on undisciplined pass-rush penalties) and applies it to all 32 NFL teams for the 2025 regular season. League-wide, defenses gained 8,336 yards on 1,287 sacks and surrendered 2,295 yards on three penalty types, a give-back rate of 27.5%. Net Sack Yards averaged 188.8 per team (SD 63.7). Rankings by the new metric track sack yardage closely (r = 0.91) but diverge sharply for undisciplined teams: Denver led the league in sacks (68) and sack yards (424) yet ranked fourth on the net measure, and Chicago, with a middling 35 sacks, finished last after surrendering 154 penalty yards. Because the penalty categories are only a proxy for pass-rush indiscipline, the results are descriptive, not causal.

1. Introduction

The sack is the most visible product of a pass rush, and sack counts and sack yards are standard defensive statistics. Yet a rush that produces sacks can also produce roughing-the-passer flags and pre-snap fouls that move the ball the other way. A defense that gains 6.5 yards per sack and gives back 15 yards on a single roughing call has not netted what the sack column suggests. This paper asks three questions about the 2025 regular season:

- How large is the give-back from undisciplined pass-rush penalties relative to sack yardage?
- How much does the ranking of teams change when penalties are netted out?
- Is pass-rush indiscipline associated with pass-rush volume?

2. Metric definition

For a defense:

> ***Net Sack Yards = yards lost by opposing offenses on sacks - yards surrendered on undisciplined pass-rush penalties***

The offense-side mirror is **Net Sack Yards Lost** = sack yards lost - penalty yards gained from opponents' pass-rush fouls (lower is better). Because every flag has one offender and one beneficiary, the league totals of the two views are identical (6,041 net yards), which provides an internal consistency check.

Penalty categories

Three accepted-penalty types stand in for "undisciplined pass rush": roughing the passer (15 yards in most cases), defensive offside, and neutral zone infraction (typically 5 yards each). Yardage is as marched, so half-the-distance and short-spot penalties are reflected.

3. Data and method

- **Sacks and sack yards (defense):** FOX Sports team defensive statistics, 2025 regular season. Sack counts were cross-checked against NFL.com for all 32 teams.
- **Sacks taken and yards lost (offense):** NFL.com team statistics, 2025 regular season. League totals match the defensive figures exactly (1,287 sacks; 8,336 yards).
- **Penalties by type and team:** NFLPenalties.com, which includes postseason games. Postseason flags were identified in each penalty log and removed: roughing the passer (Philadelphia, LA Rams), defensive offside (Buffalo twice, LA Rams, Seattle twice) and neutral zone infraction (Denver, LA Rams). After adjustment, league totals reconcile to the regular-season totals implied by the site.
- **Games:** 17 for every team. Results are season totals; per-game figures are simple division.

All analysis is descriptive. No data from other seasons are used.

4. Results

4.1 League-wide picture

Measure	League total	Per team (mean)
Sacks	1,287	40.2
Sack yards gained	8,336	260.5
Roughing the passer (yds)	1,251 (92 flags)	39.1
Defensive offside (yds)	672 (141 flags)	21.0
Neutral zone infraction (yds)	372 (75 flags)	11.6
Penalty yards surrendered	**2,295**	**71.7**
Net Sack Yards	**6,041**	**188.8**

Sacks averaged 6.5 yards. Penalties erased 27.5% of sack yardage league-wide. Roughing the passer accounted for 54.5% of penalty yards, defensive offside for 29.3% and neutral zone infraction for 16.2%, so the costliest category by yards was the 15-yard roughing call, even though it is far less frequent than defensive offside.

4.2 Team rankings

The four best defenses by Net Sack Yards were Cleveland (321), Atlanta (320), Houston (310) and Denver (302). At the other end were Chicago (50), Kansas City (75), the NY Jets (115) and San Francisco (133). The median team netted 184.5 yards. Three patterns stand out:

- **Discipline separates close contemporaries.** Cleveland (377 sack yards) and Atlanta (366) lead largely because they surrendered only 56 and 46 penalty yards. Houston matched Atlanta's sack yardage (366) while giving back 56.
- **Volume is not enough.** Denver produced the most sacks and sack yards in the league (68; 424) but gave back 122 penalty yards, the second most, dropping it from first to fourth. Minnesota (327 sack yards) netted 215 after surrendering 112.

- **One team is an outlier.** Chicago surrendered 154 penalty yards, 126 of them on nine roughing-the-passer flags, equal to 75% of its sack yards (204). Kansas City ranked second-lowest largely because it gained only 158 sack yards while surrendering 83.

4.3 Relationships between measures

Pair	Pearson r
Sack yards vs. Net Sack Yards	0.91
Sacks vs. Net Sack Yards	0.80
Sacks vs. penalty yards surrendered	0.39

Net Sack Yards is dominated by the sack-yardage term, as expected, because sack yards (mean 260.5) are several times larger than penalty yards (mean 71.7). The Spearman rank correlation between sacks and net yards is 0.88, so most teams keep a similar position. The moderate positive correlation between sacks and penalty yards ($r = 0.39$) suggests that more aggressive rushes tend to draw more flags, but with only 32 observations and a wide spread (penalty SD 27.3) it should be treated as suggestive. It does not show that rush volume causes penalties.

4.4 Offense view

Seen from the offense, the metric measures how much net yardage a team loses to sacks after crediting yards gained from opponents' flags. The best offenses were the NY Giants (48), Atlanta (66), Dallas (74) and Denver (75); the worst were Tennessee (388), Las Vegas (383), Arizona (350) and the NY Jets (321). The Giants offset 229 sack yards lost with 181 penalty yards gained, the largest benefit in the league, while Tennessee gained only 29. Across offenses, sacks taken and penalty yards gained were essentially unrelated ($r = 0.12$), which suggests that drawing pass-rush flags is not a consequence of being sacked often.

5. Discussion

Net Sack Yards reframes the pass rush as a ledger. Its correlation with raw sack yards is high, so the metric does not overturn the conventional picture, but it identifies the teams where that picture is misleading. Denver's league-best pass rush was less dominant than its raw totals imply, and Chicago's defense was a net negative on this measure despite an average sack total.

A practical reading is that roughing the passer is the lever with the most yardage at stake. It is both the heaviest penalty and the only one of the three tied directly to contact with the passer, so it is the category most plausibly related to rush technique rather than pre-snap focus.

6. Limitations

- **Proxy penalties.** Defensive offside and neutral zone infraction are pre-snap fouls that can occur on run plays, and defensive offside includes some special-teams flags. Neither source tags penalties to pass-rush snaps.
- **Omitted penalty types.** Encroachment, running into the passer, unnecessary roughness on the quarterback and face-mask calls were not available by team and are excluded.

- **Unvalued costs.** A roughing-the-passer call carries an automatic first down, which has value beyond its yardage but is not priced here.
- **No exposure adjustment.** Opponent dropbacks were not available, so team totals are not rate-adjusted. Teams that face more passes have more chances to sack and to foul.
- **Single season.** Penalty rates are volatile year to year, and one season cannot separate persistent indiscipline from variance.
- **Small sample.** Thirty-two team observations make correlation estimates imprecise.

7. Conclusion

In 2025 the league's defenses gave back more than a quarter of their sack yardage through three kinds of pass-rush penalty. Netting those yards out leaves rankings broadly intact but moves several prominent teams, most notably Denver down from first and Chicago to the bottom, and rewards disciplined units such as Cleveland, Atlanta and Houston. Future work should add dropback exposure, additional penalty types and multiple seasons to test whether pass-rush indiscipline is a stable team trait.

Data and reproducibility

The full dataset, formulas and adjustment notes are in the accompanying workbook (net_sack_yards_2025_real_data.xlsx), and the interactive view is in the Net Sack Yards dashboard. Sources: FOX Sports (defensive sacks and yards), NFL.com (offensive sacks taken and yards lost; sack-count cross-check) and NFLPenalties.com (penalties by type and team).